

Workshop No. 3 – Kidney Blood Vessel Simulation

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Introduction

This report documents the development and findings of a **cellular automaton (CA)** simulation for kidney blood vessel dynamics, completing the systems engineering pipeline from our previous workshops:

- **Workshop I:** Analyzed the *HuBMAP* dataset, identifying key challenges in vascular segmentation (biological variability, annotation ambiguity, and chaotic sensitivity to initial conditions).
- **Workshop II:** Designed a CNN-based system with ResNet50 and CBAM for histological image classification, emphasizing robustness to noise and spatial coherence.

The simulation bridges *theoretical design* and *practical implementation* by modeling how blood vessels grow, propagate, and respond to damage—aligning with the competition’s goal of understanding vascular arrangement in kidney tissues. Key innovations include:

- Integration of histological images via pixel-intensity classification.
- Pressure-driven vessel propagation rules.
- Pathological state transitions (e.g., aneurysm formation).

Cellular Automaton Design

Scenario Definition

The CA (cellular automaton) simulates microvascular dynamics in kidney tissue, addressing three key scenarios from the HuBMAP competition:

- **Normal Vasculature Growth:**
 - Simulates vessel propagation from arterial sources
 - Tests if generated networks match histological patterns
- **Pathological Responses:**
 - Models aneurysm formation/rupture cascades
 - Evaluates system stability under chaotic perturbations
- **Image-Driven Validation:**
 - Processes real histological images into the cellular automaton initial states
 - Compares simulation outcomes with ground truth annotations

Overall Architecture

Cell Types and Behaviors

Cell Type	Code	Color	Behavior
Artery	A	Red	Constantly generates vessels in cardinal directions ($\uparrow \downarrow \leftarrow \rightarrow$). Always maintains adjacent vessels.
Vessel	V	Blue (intensity based on neighbors)	Propagates to empty adjacent cells with probability <code>PROPAGATION_CHANCE</code> . Dies if isolated . Becomes aneurysm if $\geq \text{ANEURYSM_THRESHOLD}$ vessel/artery neighbors.
Aneurysm	X	Orange	Explodes if $\geq \text{ANEURYSM_EXPLODE_NEIGHBORS}$ aneurysm neighbors (cardinal). Destroys nearby vessels within <code>EXPLOSION_RADIUS</code> . Cures if surrounded by $< \text{ANEURYSM_CURE_NEIGHBORS}$ vessels.
Dead Cell	D	Black	Revives into vessel if $\geq \text{REVIVE_DEAD_NEIGHBORS}$ vessel/artery neighbors.
Empty	T	White	Can be converted into vessels by arteries or through propagation.
Glomerulus	G	Green	Static biological filters. Marked as G when well vascularized. Becomes GF (failing) if poorly irrigated (low vessel/artery support).
Glomerulus (Failing)	GF	Olive	Degrades from G if lacking support from nearby vessels or arteries.

Core Components

Component	Implementation
Grid Management	60×60 grid (10px cells)
Visualization	Tkinter canvas with dynamic coloring
Image Integration	PIL-based pixel classification

Key Parameters

Parameter	Value	Biological Meaning
<code>PROPAGATION_CHANCE</code>	0.3	Vessel growth probability
<code>ANEURYSM_THRESHOLD</code>	5	Neighbors to trigger aneurysm
<code>ANEURYSM_EXPLODE_NEIGHBORS</code>	2	Aneurysm neighbors needed to explode
<code>REVIVE_DEAD_NEIGHBORS</code>	1	Neighbors needed to revive a dead cell
<code>ANEURYSM_CURE_NEIGHBORS</code>	4	Vessels needed to heal an aneurysm
<code>EXPLOSION_RADIUS</code>	2	Region of vascular compromise due to aneurysmal burst

Systems Engineering Considerations

The simulation integrates core principles of systems engineering to ensure robustness and efficiency. The following aspects were considered during the system's design and implementation:

- **Constraints:**

- Discrete 2D grid with size defined by `GRID_SIZE`.
- Real-time rendering and interactivity using Tkinter GUI.
- Fixed time-step update logic per simulation step.

- **Resource Limits:**

- Memory usage grows quadratically with grid size: $\mathcal{O}(n^2)$ for `grid`, `pressure`, and rendering buffers.
- CPU load increases with both grid size and rule complexity.
- Tkinter’s rendering becomes visibly slower for grids larger than 80×80 .

- **Performance–Accuracy Tradeoff:**

The accuracy of the vascular behavior model improves with larger grid sizes, as more spatial resolution allows finer granularity in vessel formation and aneurysm behavior. **However, increasing `GRID_SIZE` significantly impacts performance.** A larger grid results in:

- Increased simulation step time (latency).
- Slower user interaction response.
- Higher memory and CPU usage.

Therefore, `GRID_SIZE` should be selected based on the available system resources and the level of anatomical detail required.

Prototype Code

The following code stubs represent the real structure of the simulator provided. Each section corresponds to a functional module involved in data ingestion, logic processing, or inference.

```
# === CLASSIFICATION FUNCTION ===
# Converts grayscale values to biological cell types
def classify_pixel(value):
    if value < 90: return "A" # Artery
    elif value < 130: return "V" # Vessel
    elif 130 <= value < 140: return "G" # Glomerulus
    else: return "T" # Empty tissue

# === IMAGE INGESTION CLASS ===
# Displays popup to load and convert an image to a cell grid
class ImageLoaderPopup:
    def load_image():
        # Open grayscale image
        # Resize to GRID_SIZE
        # Map each pixel using classify_pixel
        self.grid_data = [[classify_pixel(img[x, y]) for x] for y]

    def confirm_image():
        # Send classified grid back to simulator

# === MAIN SIMULATOR CLASS ===
# Manages UI, canvas, grid state and step logic
class BloodVesselSimulation:
    def __init__(self):
```

```

    # Initialize empty grid
    # Create UI toolbar
    # Bind mouse events
    # Load image popup

def set_tool(tool_code):
    # Set current drawing tool: A, V, D, X, G, etc.

def handle_click(event):
    # Update selected cell in grid
    # Draw artery or vessel manually if in edit mode

def draw_grid():
    # For each cell, draw a colored rectangle
    # Color depends on cell type and neighbor count

# === SIMULATION LOGIC ===
def step():
    # Step is called to update the grid one time unit

    # 1. Arteries generate vessels around them
    for each A in grid:
        convert adjacent T or D cells to V

    # 2. Vessels check their neighborhood
    for each V:
        if no cardinal neighbors: die (become D)
        if too many total neighbors: become aneurysm X
        else: possibly propagate to empty neighbor

    # 3. Aneurysms check for explosion or healing
    for each X:
        if 2+ X neighbors: explode and destroy nearby V
        if < 4 V neighbors: heal to V

    # 4. Dead vessels may revive
    for each D:
        if 1 V or A neighbors: become V

    # 5. Glomeruli are marked as failing or healthy
    for each G or GF:
        if well vascularized: mark as G
        else: mark as GF

    # Final step: redraw canvas with updated state

# === SUPPORT FUNCTIONS ===
def neighbors_xy(i, j):
    # Return N, S, E, W positions

def neighbors_all(i, j):
    # Return all 8 surrounding positions

```

```

def distance(x1, y1, x2, y2):
    # Euclidean distance between two cells

# === PROGRAM ENTRY POINT ===
# Starts the graphical interface
if __name__ == "__main__":
    root = Tk()
    app = BloodVesselSimulation(root)
    root.mainloop()

```

Chaos Analysis of the Cellular Automaton System

Emergent Complexity

Cellular automata (CA) can produce highly complex global behavior from a small set of local rules. This property makes them well-suited for modeling biological systems like blood vessel networks, where simple interactions lead to emergent vascular structures that are hard to predict.

Sensitivity to Initial Conditions

The system's behavior depends strongly on its initial configuration. Small differences in the placement of arteries, vessels, or glomeruli can result in vastly different outcomes, from healthy branching to early aneurysm formation. This reflects the concept of deterministic chaos.

Nonlinear Feedback and Local Rules

Each cell responds to its neighbors through nonlinear interaction rules. These include vessel propagation, aneurysm detection, and healing based on thresholds. These local rules create feedback loops that can amplify or suppress behavior, depending on the current state of the surrounding cells.

Parameter Control and Transitions

System parameters, such as propagation probability or aneurysm thresholds, act as control variables. Small adjustments in these values can cause the CA to transition between stable and unstable regimes, allowing simulation of both healthy and pathological vascular states.

Robustness Through Chaotic Exploration

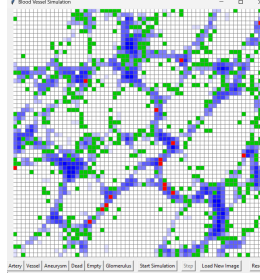
Incorporating chaotic behavior enhances the robustness and realism of the model. Exploring different parameter ranges and initial states helps reveal the boundaries of stability, ensuring the system can simulate a wide spectrum of biological responses under uncertain or noisy conditions.

Cellular Automaton Execution Under Varying Initial Conditions

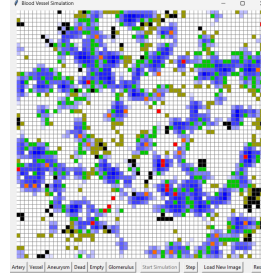
In this section, we analyze the system's sensitivity to initial conditions by comparing two distinct histological images. Each image serves as the starting state for the cellular automaton and is evolved independently through 25 simulation iterations. This comparison highlights how subtle differences in the initial vascular distribution can lead to significantly different outcomes in the resulting vascular patterns and pathological developments.



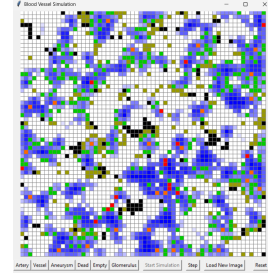
(a) Case 1 - Input



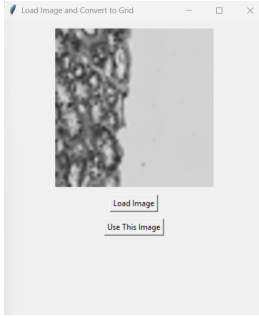
(b) Iteration #1



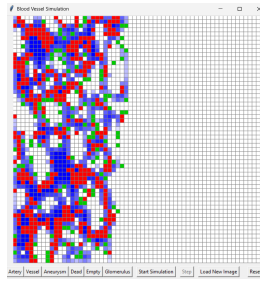
(c) Iteration #15



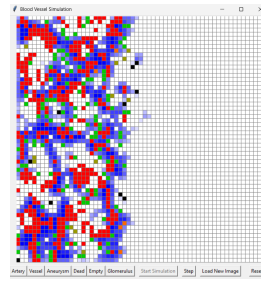
(d) Iteration #25



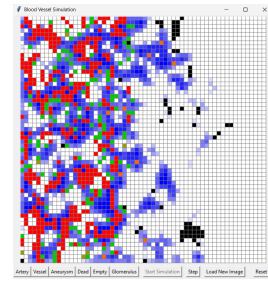
(e) Case 2 - Input



(f) Iteration #1



(g) Iteration #15

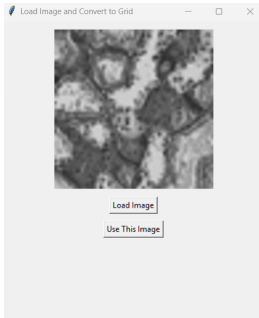


(h) Iteration #25

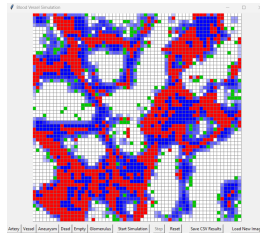
Effect of Arterial Distribution on System Stability

As observed, the initial conditions in the first input image lead to significant instability in the system. Based on our hypothesis, this is primarily due to the lack of arterial sources (red cells). The absence of consistent vessel (blue cells) generation causes the system to behave chaotically over time, failing to sustain organized vascular structures.

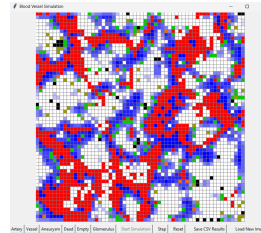
In contrast, the second input image includes a more robust and well-distributed network of arteries (red cells), providing a steady supply of blood vessels (blue cells) throughout the simulation. As a result, the vascular architecture remains more coherent and stable even after 25 iterations.



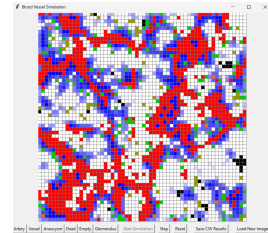
(a) Case 3 - Input



(b) Iteration #1



(c) Iteration #50



(d) Iteration #100

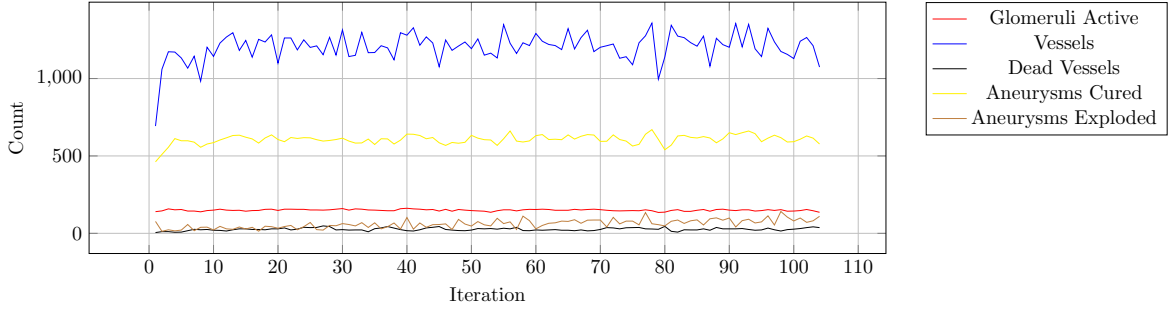


Figure 1: Simulation results over time from Case 3

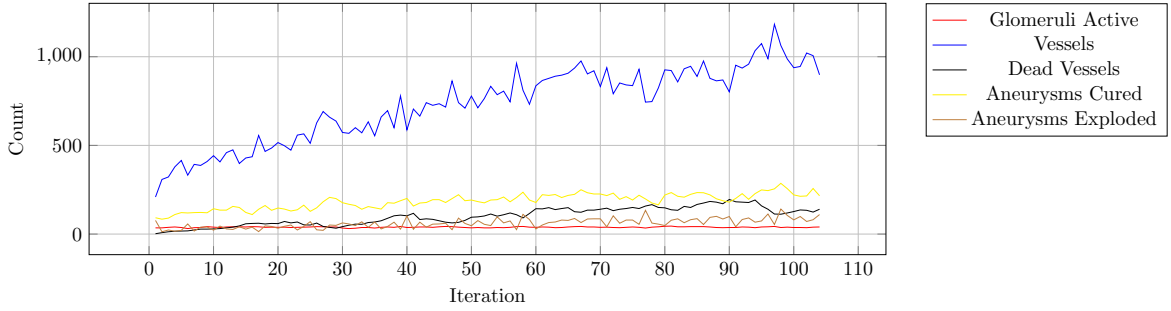
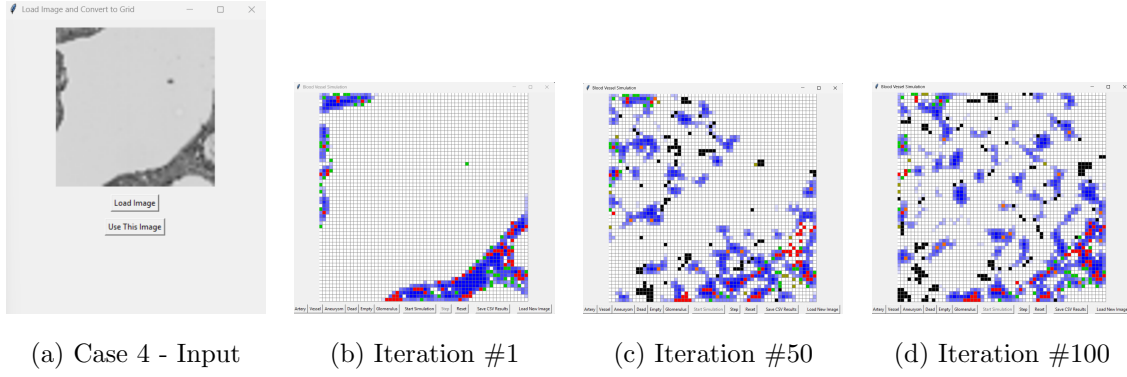


Figure 2: Simulation results over time from Case 4

Case Comparison: Arterial Structure and System Stability

In analyzing the behavior of the system across different cases, a clear relationship emerges between the initial arterial configuration and the vascular dynamics throughout the simulation. Below are the key observations comparing Case 3 and Case 4:

- **Case 3 — Structured Arterial Input:**
 - The presence of a well-organized arterial network leads to a stable generation of blood vessels (blue cells).

- The number of cured aneurysms remains steady across iterations, indicating effective vascular regulation.
- Arterial pressure remains within expected levels, reflecting a balanced supply-demand relationship within the system.
- The architecture supports vascular coherence and prevents systemic instability.

• **Case 4 — Poorly Structured Arterial Input:**

- The initial disorganized artery distribution results in erratic and insufficient vessel generation.
- As iterations progress, arterial pressure increases continuously, suggesting an accumulation of pressure due to inadequate distribution of vascular flow.
- Despite the overall rise in pressure, the system paradoxically exhibits a higher number of dead blood vessels caused by local hypoperfusion (low pressure), highlighting irregular and unstable perfusion zones.
- The system fails to maintain vascular homeostasis, and the architecture degrades progressively.

These findings reinforce the hypothesis that the spatial distribution and organization of arterial sources at the initial stage are critical determinants of long-term vascular performance and stability in the simulation environment.

Simulation Insights and Design Reflections

Key Findings and Summaries

The simulation reveals several emergent behaviors:

- In grids with a dense arterial structure, blood vessels propagate smoothly, and glomeruli remain functional.
- In images with sparse or inconsistent arterial input, chaotic patterns emerge, including frequent aneurysm formation and cell death.
- Aneurysms tend to cluster in overpopulated vessel zones, often leading to chain reactions of damage (rupture and degeneration).

Simulation steps can be logged quantitatively using:

- Vessel count over time.
- Aneurysm frequency and survival rate.
- Glomerulus function status (**G** vs **GF**) as a proxy for overall kidney health.

Discrepancies and Observations

While the design assumed a stable spread of vessels from arterial origins, the simulation showed that:

- Vessels can stagnate or regress if arterial support is insufficient.
- Aneurysm thresholds are highly sensitive; a single extra neighbor can destabilize entire zones.
- Dead zones can revive, but only under narrow conditions—contrary to the expectation of broader regeneration.

Conclusions

The cellular automaton developed for simulating renal blood vessel dynamics successfully captures the nonlinear, emergent behavior of vascular systems under both normal and pathological conditions. Through iterative modeling, the simulation demonstrates a strong dependency on initial arterial configuration and local interaction rules.

- **Initial arterial structure determines system stability:** Well-distributed arterial sources promote steady vessel growth and prevent chaotic outcomes such as uncontrolled aneurysm formation.
- **Chaotic sensitivity enhances biological realism:** Small changes in initial input images lead to divergent vascular outcomes, mirroring real-world biological variability and uncertainty.
- **Vascular resilience requires local feedback:** Systems with consistent vessel support near glomeruli maintain higher glomerular function and fewer aneurysmal events, confirming the importance of micro-scale feedback.
- **Pathological zones self-amplify:** Once aneurysms or dead vessels dominate a region, the system tends to degenerate unless strong arterial reinforcement is introduced.
- **Model parameters offer tunability:** Thresholds and probabilities governing propagation, rupture, and healing provide a rich space for simulating diverse pathological scenarios.

These results highlight the potential of cellular automata as a tool for exploring vascular phenomena and underscore the need for precise initial condition design in computational models of biological tissue.